

CS5340

Uncertainty Modelling in AI

Lecture 1: Introduction to Probabilistic Reasonings

Assoc. Prof. Lee Gim Hee

AY 2026/27

Semester 1

Course Schedule

Week	Date	Topic	Remarks
1	10 Aug	National Day Holiday	No Lecture
2	17 Aug	Introduction to probabilistic reasoning	Assignment 0: Python Numpy Tutorial (Ungraded)
	22 Aug	Bayesian networks (Directed graphical models)	Make-up Lecture: Sat, 9.30am-12.30pm (LT15, Recording will be provided)
3	24 Aug	Markov random Fields (Undirected graphical models)	
4	31 Aug	Variable elimination and belief propagation	Assignment 1: Belief propagation and maximal probability (15%) Assignment 1: Due (13 Sep, 2359 HRS)
	05 Sept	Factor graph and the junction tree algorithm	Make-up Lecture: Sat, 9.30am-12.30pm (LT15, Recording will be provided)
5	07 Sep	I will be traveling	No Lecture
6	14 Sep	Parameter learning with complete data	Assignment 2: Junction tree and parameter learning (15%) Assignment 2: Due (27 Sep, 2359 HRS)
-	21 Sep	Recess week	No Lecture
7	28 Sep	Mixture models and the EM algorithm	
8	05 Oct	Hidden Markov Models (HMM)	Assignment 3: Hidden Markov model (15%) Assignment 3: Due (18 Oct, 2359 HRS)
9	12 Oct	Monte Carlo inference (Sampling)	
10	19 Oct	Variational inference	Assignment 4: MCMC Sampling (15%) Assignment 4: Due (01 Nov, 2359 HRS)
11	26 Oct	Variational Autoencoder and Diffusion Models	
12	02 Nov	Graph-cut and alpha expansion	
13	09 Nov	Diwali/Deepavali	No Lecture

Final Exam: 21 Nov 2026

Acknowledgements

- A lot of slides and content of this lecture are adopted from:
 1. Simon Prince, “Computer Vision: Models, Learning, and Inference”, Chapter 1 and 2.
 2. Daphne Koller and Nir Friedman, "Probabilistic graphical models", Chapter 2.
 3. Christopher Bishop, “Pattern Recognition and Machine Learning”, Chapter 2.

Learning Outcomes

Students should be able to:

1. Describe uncertain quantities with **random variables** and **joint probabilities**.
2. Explain the basic rules of probability – **sum**, **product**, **Bayes'**, **independence** and **expectation** rules.
3. Use the common probabilities distributions – **Bernoulli**, **categorical**, **univariate** and **multivariate normal** distributions.
4. Explain the use of **conjugate distributions**.

Probability Space

- A probability space (Ω, E, P) models a process consisting of outcomes that occur **randomly**.
- Consists of **three parts**:
 1. Outcome or sample space Ω
 2. Event space E
 3. Probability distribution $P: E \rightarrow \mathbb{R}$

Outcome/Sample and Event Spaces

- Outcome/sample space is an agreed upon **set of possible outcomes**, denoted by Ω .
- Event space $E \subseteq 2^\Omega$ is a **subset of the power set** of Ω , it is the set of **measurable events** to which we assign probabilities.

Outcome and Event Spaces

- Event space must satisfy **three basis properties**:
 1. It **must contain** the **empty event** ϕ , and the **trivial event** Ω .
 2. It is **closed under countable unions**, i.e. if $\alpha_i \in E \ \forall i = 1, 2, \dots$, then so is $\bigcup_{i=1}^{\infty} \alpha_i$.
 3. It is **closed under complements**, i.e. if $\alpha \in E$, then so is $\Omega - \alpha$.

Outcome/Sample and Event Spaces

Example 1:

Let's consider a 6-faced dice. The **outcome/sample space** is given by $\Omega = \{1, 2, 3, 4, 5, 6\}$.

A possible **event space** is $E = \{\{1,3,5\}, \{2,4,6\}, \emptyset, \{1,2,3,4,5,6\}\}$, i.e. event of a throw is even or odd.

Check:

1. E contains the **empty** $\emptyset$ and **trivial** $\{1,2,3,4,5,6\}$ sets.
2. Let $\alpha_1 = \{1,3,5\}$ and $\alpha_2 = \{2,4,6\}$, i.e., $\alpha_1, \alpha_2 \in E$, then $\alpha_1 \cup \alpha_2 = \{1,2,3,4,5,6\} \in E$ because E is **closed under countable unions**.
3. Let $\alpha = \{2,4,6\} \in E$, then $\{1,2,3,4,5,6\} - \{2,4,6\} = \{1,3,5\} = \{\Omega - \alpha\} \in E$ because E is **closed under complement**.

Remark: Check for yourself that 2 and 3 are always true $\forall \alpha \in E$!

Outcome/Sample and Event Spaces

Example 2:

Let's consider measuring the lifetime of a lightbulb. The **outcome/sample space** is given by $\Omega = [0, \infty)$.

A possible **event space** is $E = \{[0, 90), [90, \infty), \emptyset, [0, \infty)\}$, i.e. event of lightbulb lifespan ≥ 90 .

Check:

1. E contains the **empty** $\emptyset$ and **trivial** $[0, \infty)$ sets.
2. Let $\alpha_1 = \emptyset$ and $\alpha_2 = [0, \infty)$, i.e., $\alpha_1, \alpha_2 \in E$, then $\alpha_1 \cup \alpha_2 = [0, \infty) \in E$ because E is **closed under countable unions**.
3. Let $\alpha = [0, 90) \in E$, then $[0, \infty) - [0, 90) = [90, \infty) = \{\Omega - \alpha\} \in E$ because E is **closed under complement**.

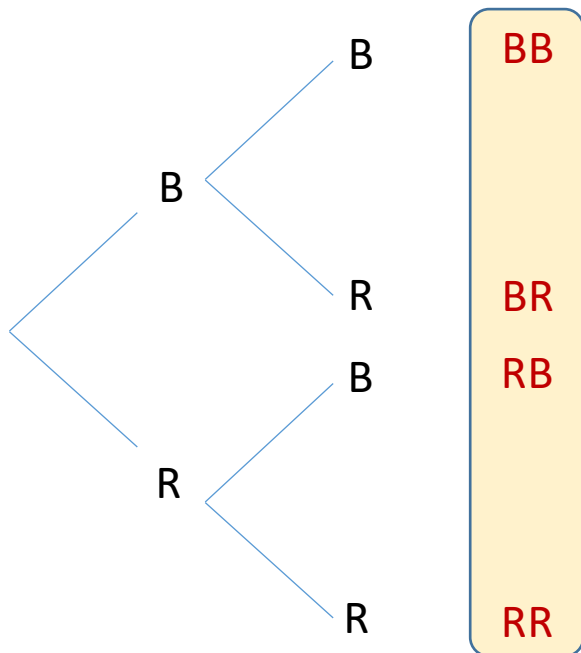
Remark: Check for yourself that 2 and 3 are always true $\forall \alpha \in E$!

Outcome/Sample and Event Spaces

Example 3:

Picking 2 marbles, one at a time, from a bag that contains many blue and red marbles. Find the sample space?

Tree Diagram



List

$\{BB, BR, RB, RR\}$

Table

	B	R
B	BB	BR
R	RB	RR

Probability Distributions

- A probability distribution P over (Ω, E) is a **mapping from events in E to real values** ($P: E \rightarrow \mathbb{R}$) that satisfies the following conditions, i.e. axioms of probability:
 1. **Non-negativity**, i.e. $P(\alpha) \geq 0, \forall \alpha \in E$.
 2. Probability of all outcomes **sums to 1**, i.e. $P(\Omega) = 1$.
 3. **Mutually disjoint events**: If $\alpha, \beta \in E$ and $\alpha \cap \beta = \emptyset$, then $P(\alpha \cup \beta) = P(\alpha) + P(\beta)$.

Random Variables

- A random variable, denoted as X (**upper case**), is the formal machinery for discussing **attributes** and their **values** in different outcomes.
- More formally: given a probability space (Ω, E, P) , a random variable is **a function** $X: \Omega \rightarrow S$ that maps a set of possible outcomes Ω to a measurable space S .
- Typically, S is the set of **real numbers**, i.e. $S \in \mathbb{R}$.

Random Variables

- The **set of values** that a random variable X can take is denoted as $Val(X)$.
- A lower-case letter, e.g. x , is used to refer to a **generic value** of a random variable X , a.k.a. **realization** of the random variable.

Example: We write $P(X = x)$ for all $x \in Val(X)$.

- $P(x)$ is often used as a **shorthand notation** for $P(X = x)$.
- We use the notation x^i to represent a **specific value** of X .

Random Variables

- The value of a random variable $Val(X)$ can be:
 - **Discrete**, i.e. takes values from a **predefined set**, or
 - **Continuous**, i.e. takes values that are **real numbers**.

Examples:

Random variables with discrete values

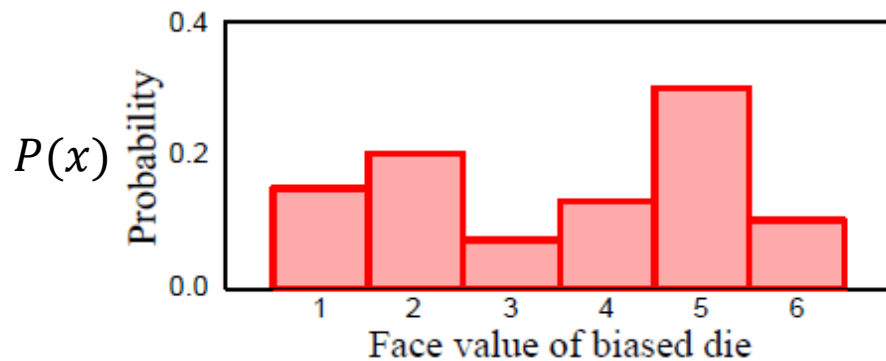
- Rolling a six-faced die: $Val(X) = \{1, 2, \dots, 6\}$
- Weather conditions: $Val(X) = \{\text{"rain"}, \text{"cloud"}, \text{"snow"}, \text{"sun"}, \text{"wind"}\}$
- Number of people on the next train: $Val(X) = \mathbb{Z}_{\geq 0}$

Continuous random variables

- Time taken to finish an exam: $Val(X) = [1, 2]$ hours
- Height of a tree: $Val(X) = \mathbb{R}_{>0}$
- Ambient Temperature: $Val(X) = \mathbb{R}$

Probability Distributions: Discrete Vs Continuous

- Discrete: **Probability mass function**, $P(x)$



$Val(X) = \{1, 2, 3, 4, 5, 6\}$

$$\sum_{i=1}^K P(X = x^i) = 1$$

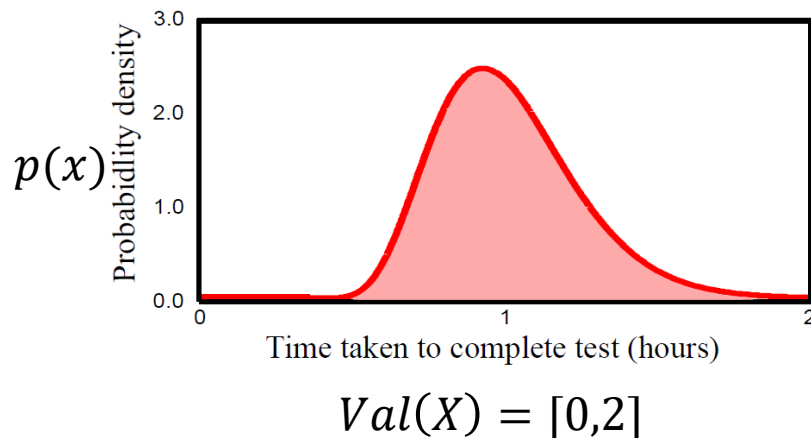
$$0 \leq P(X = x^i) \leq 1,$$

$$\forall i = 1, \dots, K,$$

$$\text{where } K = |Val(X)|$$

Probability Distributions: Discrete Vs Continuous

- Continuous: **Probability density function** is a function (denoted by a lower case p) $p(x): \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$.



$$\int_{Val(X)} p(x) dx = 1;$$

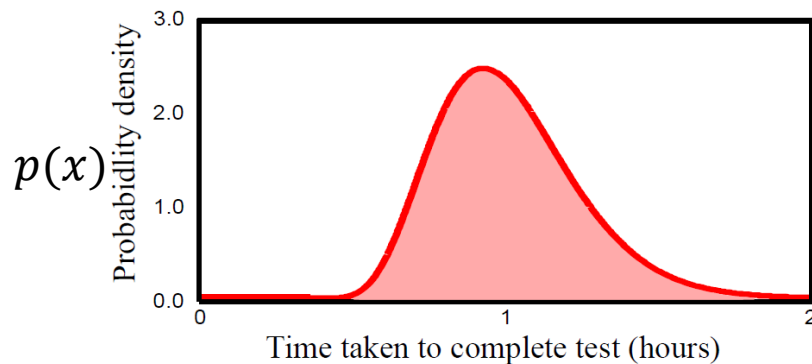
$$p(X = x^i) \geq 0, \quad \forall x^i \in Val(X)$$

Images Source: "Computer Vision: Models, Learning, and Inference", Simon Prince

Probability Distributions: Discrete Vs Continuous

- Continuous: **Probability density function** is a function (denoted by a lower case p) $p(x): \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$.

$P(X)$ is the **cumulative function** of X :



$Val(X) = [0,2]$

$$P(X \leq a) = \int_{-\infty}^a p(x) dx$$

$$P(a \leq X \leq b) = \int_a^b p(x) dx$$

$$P(X = x^i) = \int_{x^i}^{x^i} p(x) dx = 0,$$

$$\forall x^i \in Val(X)$$

Images Source: "Computer Vision: Models, Learning, and Inference", Simon Prince

Probability Distributions: Discrete Vs Continuous

In this course, we **abuse the notation** by denoting both the probability mass function and probability density function as the lower case $p(x)$!

We silently note the property differences in $P(x)$ when X is **discrete or continuous**.

Probabilistic Reasoning

Probabilistic Modeling:

- The **central paradigm** of probabilistic reasoning is to:
 1. Identify all **relevant variables** $X_1, \dots, X_N$ of the environment, and
 2. make a **probabilistic model** $p(X_1, \dots, X_N)$ of their interactions.

Probabilistic Reasoning

Probabilistic Inference:

- Reasoning (**inference**) is then performed by:
 1. Introducing **evidence** that sets variables in known state, and
 2. subsequently computing probabilities of interest, **conditioned on** this evidence.

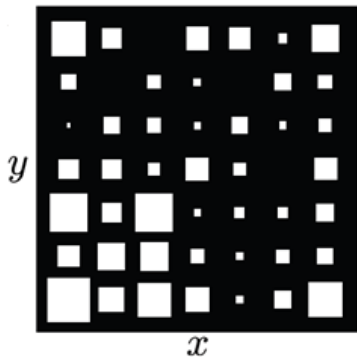
Probabilistic Reasoning

- To this end, we require the definitions of **joint probability, marginalization, conditional probability, Bayes' rule, and independence.**
- In this lecture, we look at the use of these definitions for probabilistic modeling and inference of a **small number of variables.**
- In the subsequent lectures, we will look at the use of these definitions with **graphical models** for **large number of variables.**

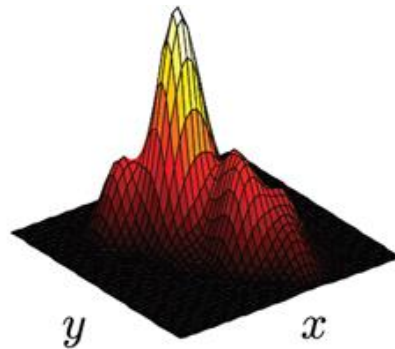
Probability: Joint Probability

- Consider **all combination** of events of two random variables X and Y .
- Some combinations of outcomes are **more likely** than others.

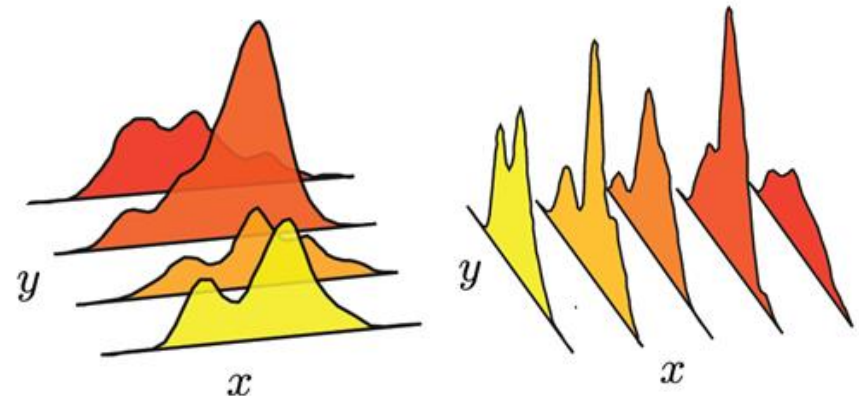
Discrete



Continuous



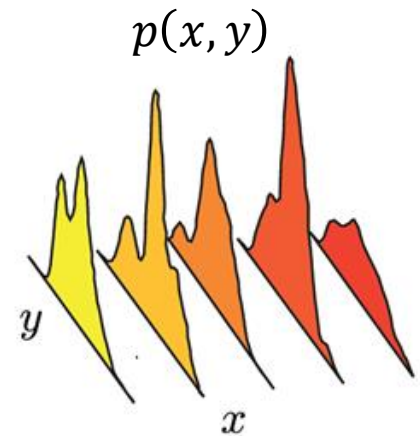
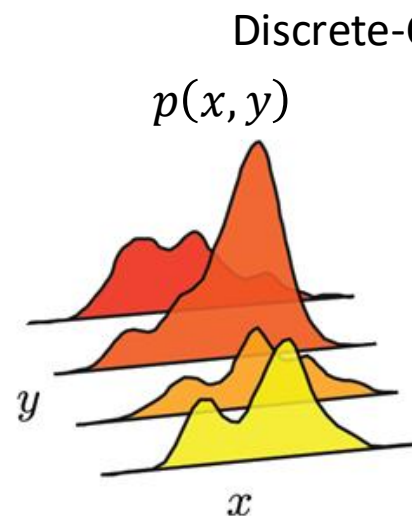
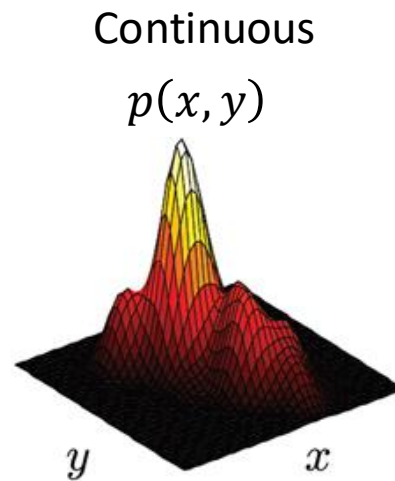
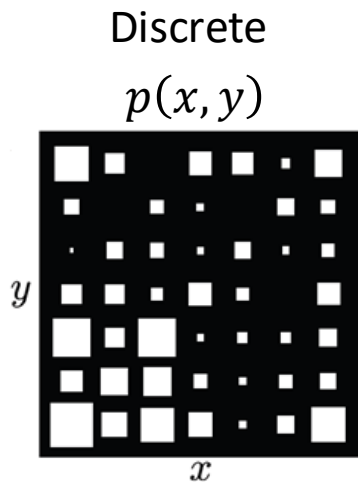
Discrete-Continuous



Images Source: "Computer Vision: Models, Learning, and Inference", Simon Prince

Probability: Joint Probability

- This is captured in the **joint probability** distribution $p(x, y)$.
- Read as “**probability of X and Y** ”.
- Can be **more than two** random variables, i.e. $p(a, b, c, \dots)$.



Images Source: “Computer Vision: Models, Learning, and Inference”, Simon Prince

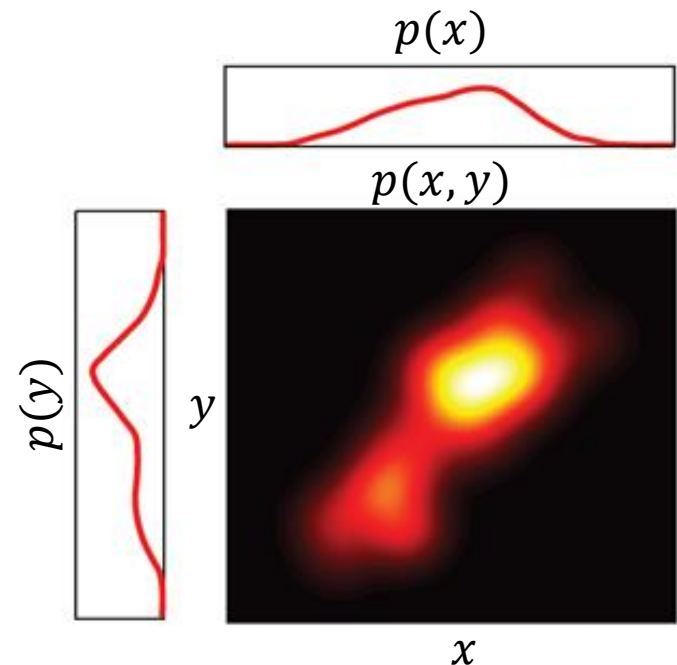
Probability: Marginalization

- Recover probability distribution of any variable in a joint distribution by **integrating (or summing)** over all other variables.
- Also known as the **“sum rule”** of probability.

Continuous:

$$p(x) = \int p(x, y) dy$$

$$p(y) = \int p(x, y) dx$$



Images Source: “Computer Vision: Models, Learning, and Inference”, Simon Prince

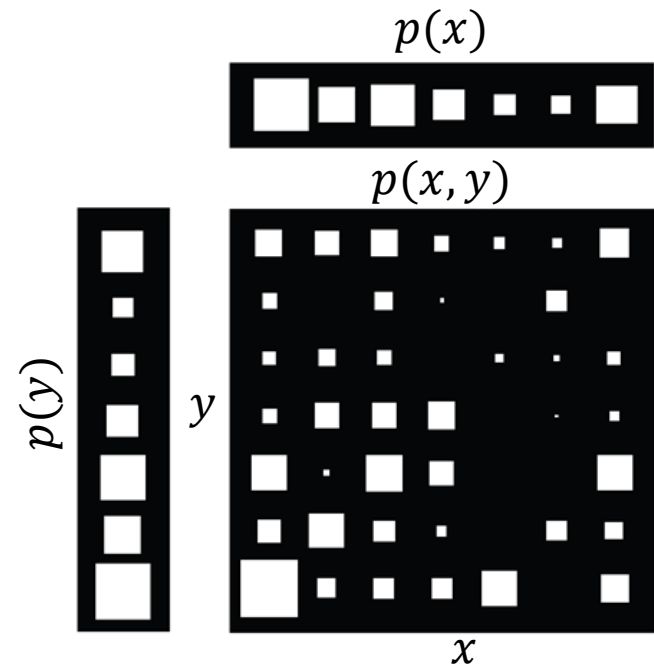
Probability: Marginalization

- Recover probability distribution of any variable in a joint distribution by **integrating (or summing)** over all other variables.
- Also known as the **“sum rule”** of probability.

Discrete:

$$p(x) = \sum_y p(x, y)$$

$$p(y) = \sum_x p(x, y)$$



Images Source: “Computer Vision: Models, Learning, and Inference”, Simon Prince

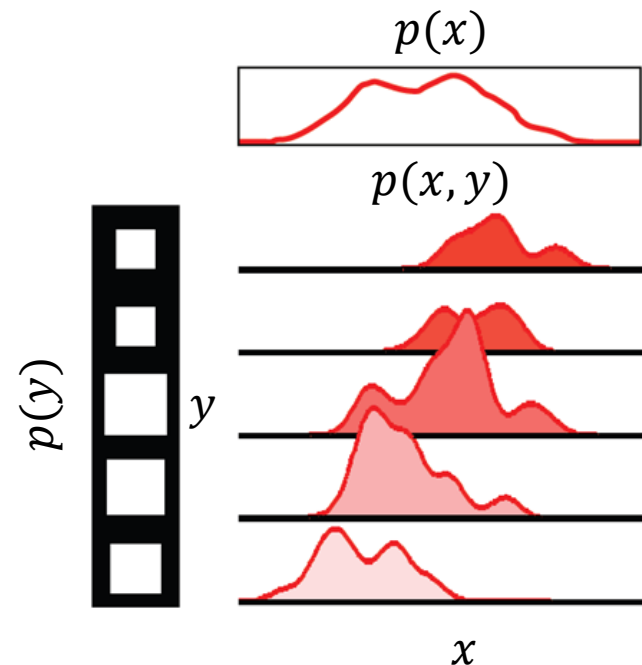
Probability: Marginalization

- Recover probability distribution of any variable in a joint distribution by **integrating (or summing)** over all other variables.
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Discrete-continuous:

$$p(x) = \sum_y p(x, y)$$

$$p(y) = \int p(x, y) dx$$



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Probability: Marginalization

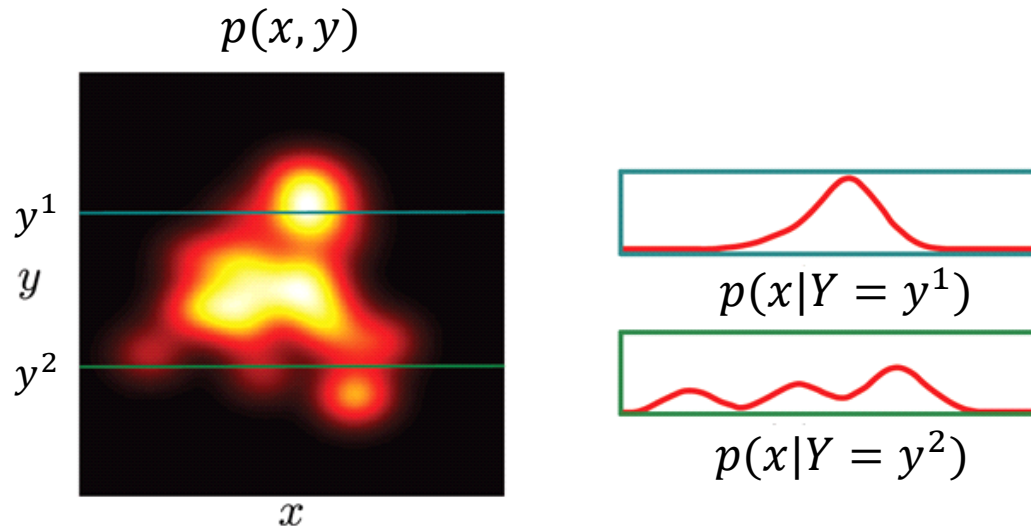
- Works in **higher dimensions** too!

Example:

$$p(x, y) = \sum_w \int p(w, x, y, z) dz$$

Probability: Conditional Probability

- $p(x|Y = y^*)$: “probability of X given $Y = y^*$ ”.
- Also known as “chain rule” or “product rule” of probability.
- **Relative propensity** of the random variable X to take different outcomes given that the random variable Y is fixed to value y^* .

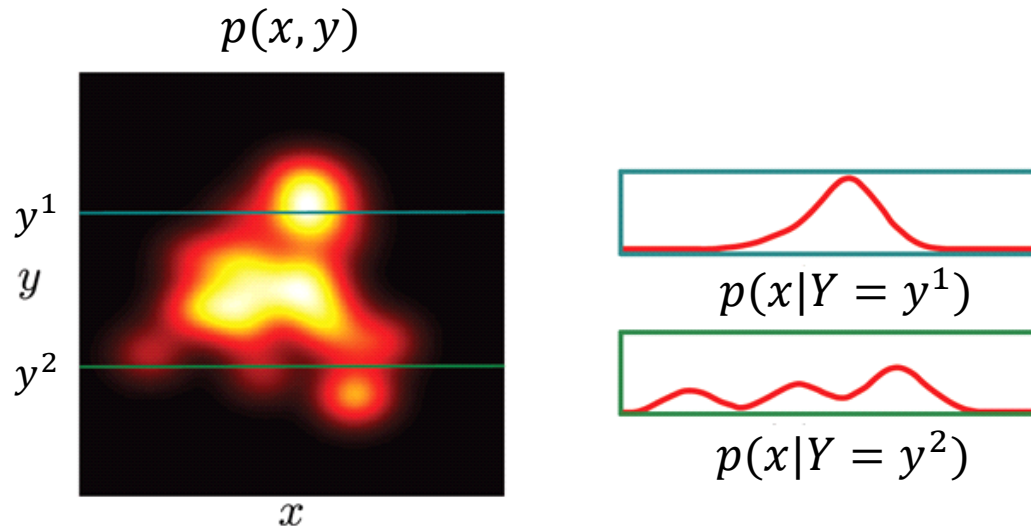


Images Source: “Computer Vision: Models, Learning, and Inference”, Simon Prince

Probability: Conditional Probability

- Conditional probability can be **extracted from joint probability**.
- Extract appropriate slice and **normalize** (so that the area is 1):

$$P(x|Y = y^*) = \frac{p(x, Y = y^*)}{\int p(x, Y = y^*)dx} = \frac{p(x, Y = y^*)}{p(Y = y^*)}$$



Images Source: "Computer Vision: Models, Learning, and Inference", Simon Prince

Probability: Conditional Probability

$$P(x|Y = y^*) = \frac{p(x, Y = y^*)}{\int p(x, Y = y^*)dx} = \frac{p(x, Y = y^*)}{p(Y = y^*)}$$

- Usually written in compact form:

$$p(x|y) = \frac{p(x, y)}{p(y)}$$

- Which can be re-arranged to give:

$$\begin{aligned} p(x, y) &= p(x|y)p(y) \\ p(x, y) &= \underbrace{p(y|x)p(x)} \end{aligned}$$

Hence, the name “product rule”!

Probability: Conditional Probability

$$p(x, y) = p(x|y)p(y)$$

- Works for **higher dimensions** too!

Example:

$$\begin{aligned} p(w, x, y, z) &= p(w, x, y|z)p(z) \\ &= p(w, x|y, z)p(y|z)p(z) \\ &= p(w|x, y, z)p(x|y, z)p(y|z)p(z) \end{aligned}$$

Probability: Bayes' Rule

- Recall:

$$p(x, y) = p(x|y)p(y)$$
$$p(x, y) = p(y|x)p(x)$$



Thomas Bayes
1701–1761

- Eliminating $p(x, y)$, we get:

$$p(y|x)p(x) = p(x|y)p(y)$$

- Rearranging:

$$p(y|x) = \frac{p(x|y)p(y)}{p(x)} = \frac{p(x|y)p(y)}{\int p(x, y)dy} = \frac{p(x|y)p(y)}{\int p(x|y)p(y)dy}$$

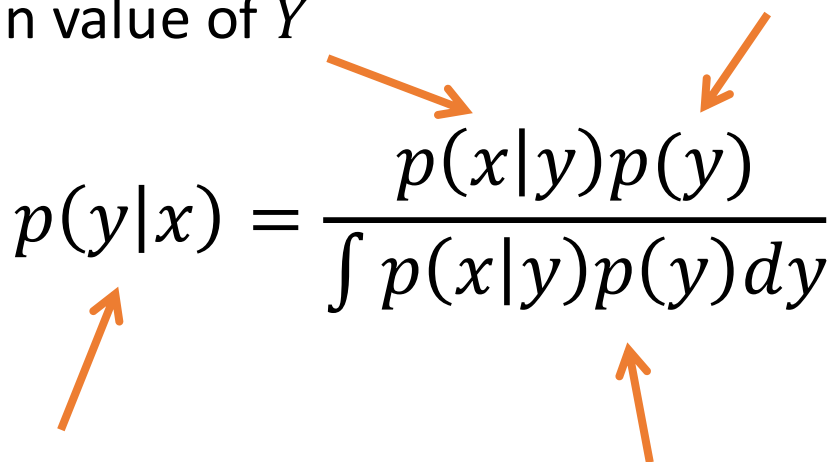
Image source: "Pattern Recognition and Machine Learning", Christopher Bishop

Probability: Bayes' Rule

Terminology:

Likelihood – propensity for observing a certain value of X given a certain value of Y

Prior – what we know about Y before seeing X


$$p(y|x) = \frac{p(x|y)p(y)}{\int p(x|y)p(y)dy}$$

Posterior – what we know about Y after observing X

Evidence – a constant to ensure that the left-hand side is a valid distribution

Probability: Example

Let random variables B and F represent the box color and type of fruit respectively, where $Val(B) = \{r, b\}$ and $Val(F) = \{a, o\}$.

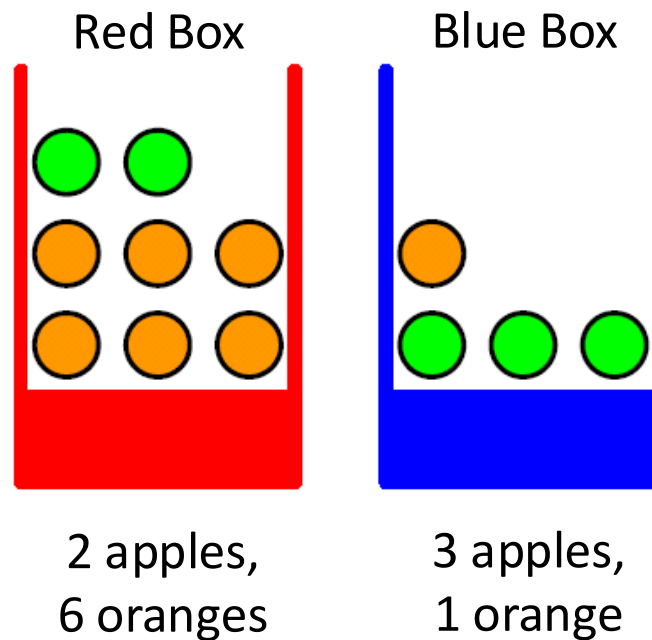


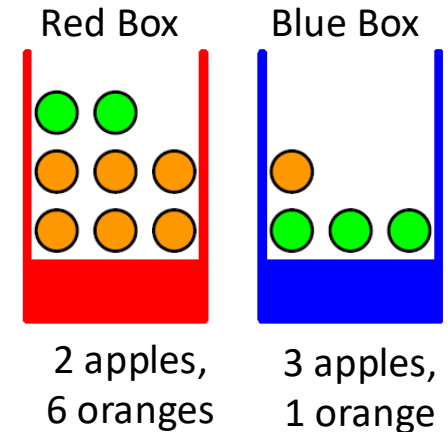
Image source: "Pattern Recognition and Machine Learning", Christopher Bishop

Probability: Example

Given:

- Probabilities of selecting either the red or the blue boxes,

$$p(B = r) = 0.4$$
$$p(B = b) = 0.6$$



- Conditional probabilities for the type of fruit, given the selected box,

$$p(F = a|B = r) = 0.25$$
$$p(F = o|B = r) = 0.75$$
$$p(F = a|B = b) = 0.75$$
$$p(F = o|B = b) = 0.25$$

Image source: "Pattern Recognition and Machine Learning", Christopher Bishop

Probability: Example

Find:

- a) The overall probability of choosing an apple.
- b) Identify the color of the box if we observed that an orange has been selected.

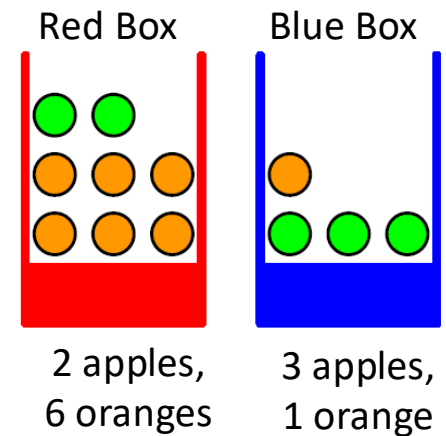


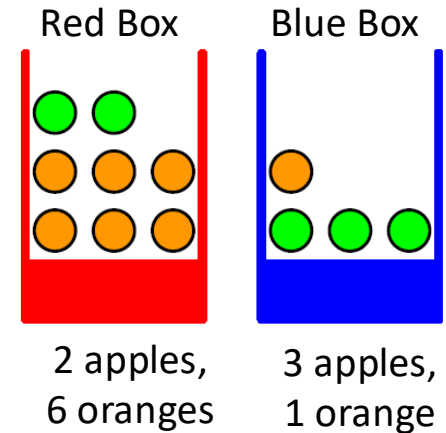
Image source: "Pattern Recognition and Machine Learning", Christopher Bishop

Probability: Example

Solution:

a) The overall probability of choosing an apple.

Using the **sum and product rules** of probability:



$$\begin{aligned} p(F = a) &= \sum_B p(F = a, B) = \sum_B p(F = a|B)p(B) \\ &= p(F = a|B = r)p(B = r) + p(F = a|B = b)p(B = b) \\ &= (0.25)(0.4) + (0.75)(0.6) = 0.55 \end{aligned}$$

Image source: "Pattern Recognition and Machine Learning", Christopher Bishop

Probability: Example

Solution:

b) Identify the color of the box if we observed that an orange has been selected.

Using **Bayes' theorem**:

$$\begin{aligned} p(B = r|F = o) &= \frac{p(F = o|B = r)p(B = r)}{p(F = o)} \\ &= \frac{p(F = o|B = r)p(B = r)}{1 - p(F = a)} = \frac{(0.75)(0.4)}{1 - 0.55} \\ &= 0.667 \end{aligned}$$

$$p(B = b|F = o) = 1 - p(B = r|F = o) = 1 - 0.667 = 0.333$$

The orange is more likely to be selected from the **red box**!

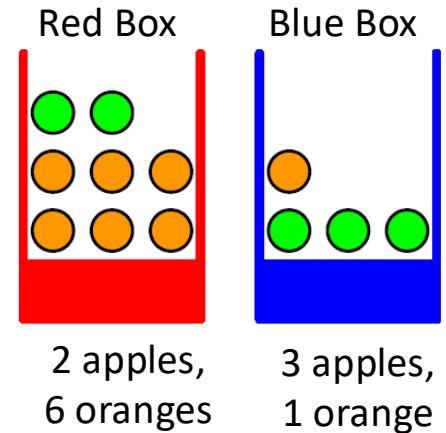


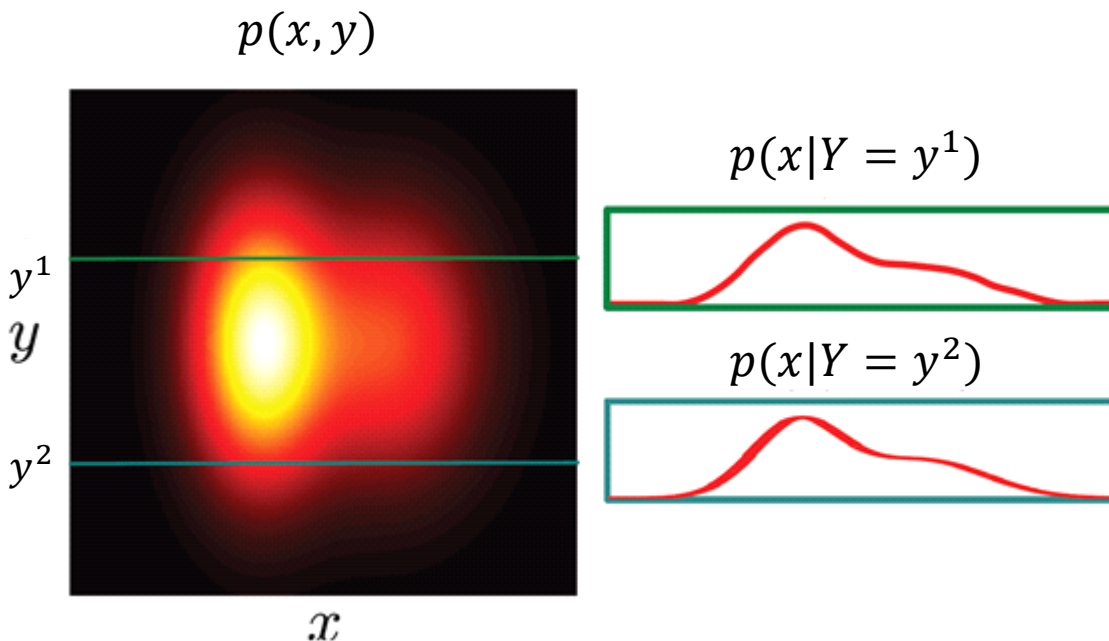
Image source: "Pattern Recognition and Machine Learning", Christopher Bishop

Probability: Independence

- The independence of X and Y means that **every conditional distribution is the same**.
- The value of Y **tells us nothing** about X and vice-versa.

$$p(x|y) = p(x)$$

$$p(y|x) = p(y)$$



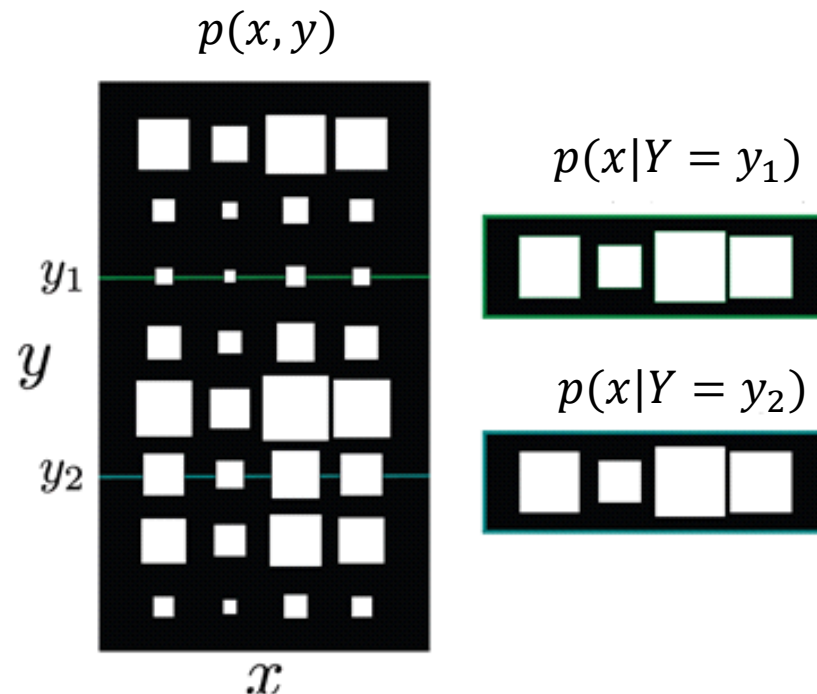
Images Source: "Computer Vision: Models, Learning, and Inference", Simon Prince

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Images Source: "Computer Vision: Models, Learning, and Inference", Simon Prince

Probability: Independence

- When variables are **independent**, the joint factorizes into a **product of the marginals**:

$$\begin{aligned} p(x, y) &= p(x|y)p(y) \\ &= p(x)p(y) \end{aligned}$$

Probability: Expectation

- The **expected or average value** of some function $f[x]$ considering the distribution of X .

Definition:

$$E[f[x]] = \sum_x f[x]p(x)$$
$$E[f[x]] = \int f[x]p(x)dx$$

Probability: Rules of Expectation

- **Rule 1:** Expected value of a **constant** is the constant.

$$E[\kappa] = \kappa$$

- **Rule 2:** Expected value of **constant times function** is constant times expected value of function.

$$E[\kappa f[x]] = \kappa E[f[x]]$$

Probability: Rules of Expectation

- **Rule 3:** Expectation of **sum of functions** is sum of expectation of functions.

$$E[f[x] + g[x]] = E[f[x]] + E[g[x]]$$

- **Rule 4:** Expectation of **product of functions in variables X and Y** is product of expectations of functions if X and Y are independent.

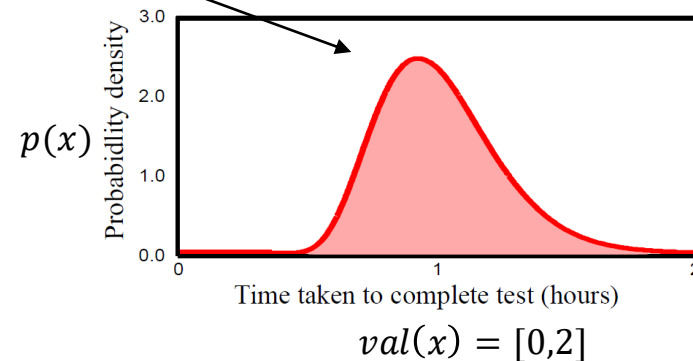
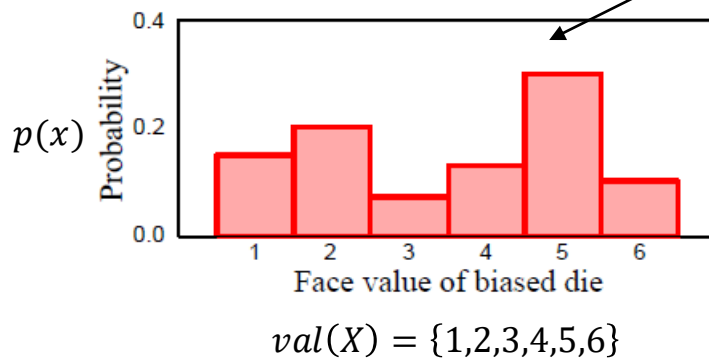
$$E[f[x]g[y]] = E[f[x]]E[g[y]],$$

if X and Y are independent

Probability Distributions

- We have seen the definitions of random variables, probability, and rules for manipulating probabilities.
- One question that remains unanswered is: “How do we assign the values of $p(x)$?”

How to get $p(x)$?



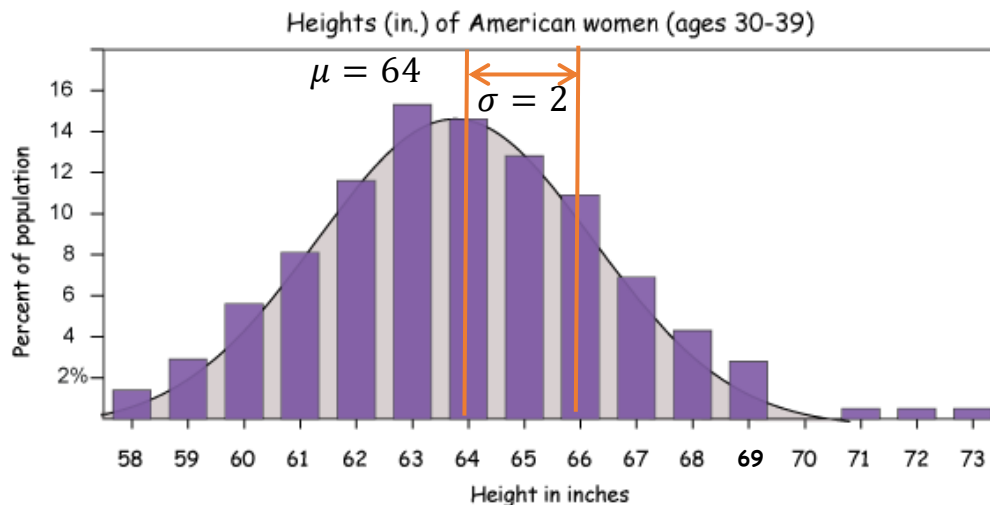
Images Source: “Computer Vision: Models, Learning, and Inference”, Simon Prince

Probability Distributions

Q: “How do we assign the probability values?”

A: Use **probability distributions** defined over some **parameters** learned from data!

Example:



Fitting a Normal distribution to the heights of a population:

$$p(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp -\frac{(x - \mu)^2}{2\sigma^2}$$

Parameters: mean $\mu = 64$, variance $\sigma^2 = 4$ are learned from data.

Image source: http://www.drcruzan.com/ProbStat_Distributions.html

Common Probability Distributions

- The choice of distribution depends on the **type/domain of data** to be modeled.

Data Type	Domain	Distribution
univariate, discrete, binary	$x \in \{0, 1\}$	Bernoulli
univariate, discrete, multi-valued	$x \in \{1, 2, \dots, K\}$	categorical
univariate, continuous, unbounded	$x \in \mathbb{R}$	univariate normal
univariate, continuous, bounded	$x \in [0, 1]$	beta
multivariate, continuous, unbounded	$\mathbf{x} \in \mathbb{R}^K$	multivariate normal
multivariate, continuous, bounded, sums to one	$\mathbf{x} = [x_1, x_2, \dots, x_K]^T$ $x_k \in [0, 1], \sum_{k=1}^K x_k = 1$	Dirichlet
bivariate, continuous, x_1 unbounded, x_2 bounded below	$\mathbf{x} = [x_1, x_2]$ $x_1 \in \mathbb{R}$ $x_2 \in \mathbb{R}^+$	normal-scaled inverse gamma
multivariate vector $\mathbf{x}$ and matrix $\mathbf{X}$, $\mathbf{x}$ unbounded, $\mathbf{X}$ square, positive definite	$\mathbf{x} \in \mathbb{R}^K$ $\mathbf{X} \in \mathbb{R}^{K \times K}$ $\mathbf{z}^T \mathbf{X} \mathbf{z} > 0 \quad \forall \mathbf{z} \in \mathbb{R}^K$	normal inverse Wishart

Bernoulli Distribution

- **Single binary** random variable X , i.e. $x \in \{0,1\}$
- A **single parameter** $\lambda \in [0,1]$.

$$\begin{aligned}p(X = 0 \mid \lambda) &= 1 - \lambda \\p(X = 1 \mid \lambda) &= \lambda\end{aligned}$$

Or

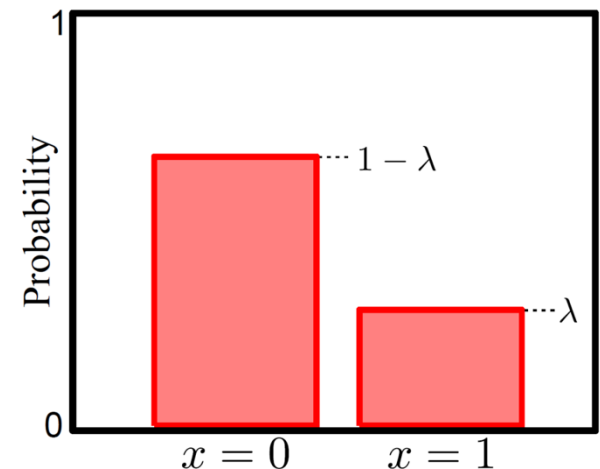
$$\begin{aligned}p(x) &= \lambda^x (1 - \lambda)^{1-x}, \\p(x) &= \text{Bern}_x[\lambda]\end{aligned}$$

Example:

X is the outcome of flipping a coin, $X = 1$ represents 'heads', and $X = 0$ represents 'tails'.



Jacob Bernoulli
1654–1705



Images source: "Pattern Recognition and Machine Learning", Christopher Bishop
"Computer Vision: Models, Learning, and Inference", Simon Prince

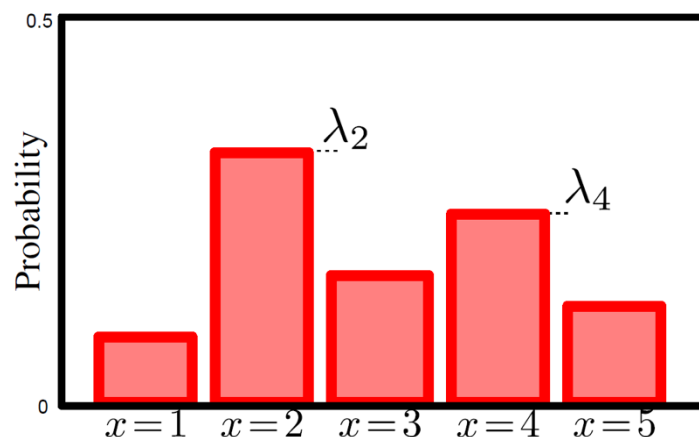
Categorical Distribution

- Discrete variables $\mathbf{X}$ that take on **1-of- K possible mutually exclusive states**, e.g. a K -faced die.
- $\mathbf{x}$ is represented by a **K -dimensional vector** $\mathbf{e}_k$ in which one of the elements $x_k = 1$, and $\sum_{k=1}^K x_k = 1$.
- e.g. $K = 5$, and $\mathbf{x} = \mathbf{e}_3 = [0, 0, 1, 0, 0]^\top$.
- K parameters** $\lambda = [\lambda_1, \dots, \lambda_K]^\top$, where $\lambda \geq 0$, $\sum_k \lambda_k = 1$.

$$p(\mathbf{X} = \mathbf{e}_k \mid \lambda) = \lambda_k$$

Or

$$p(\mathbf{x}) = \prod_{k=1}^K \lambda_k^{x_k} = \lambda_k,$$
$$p(\mathbf{x}) = \text{Cat}_x[\lambda]$$



Images Source: "Computer Vision: Models, Learning, and Inference", Simon Prince

Univariate Normal Distribution

- Also known as the **Gaussian distribution**.
- Univariate normal distribution describes **single continuous variable** X , i.e. $x \in \mathbb{R}$.
- **Two parameters** $\mu \in \mathbb{R}$ (mean) and $\sigma^2 > 0$ (variance).



Carl Friedrich Gauss
1777–1855

$$p(X = a \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp -\frac{(a-\mu)^2}{2\sigma^2}, \quad a \in \mathbb{R}$$

Or

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp -\frac{(x - \mu)^2}{2\sigma^2}$$
$$p(x) = \text{Norm}_x[\mu, \sigma^2]$$

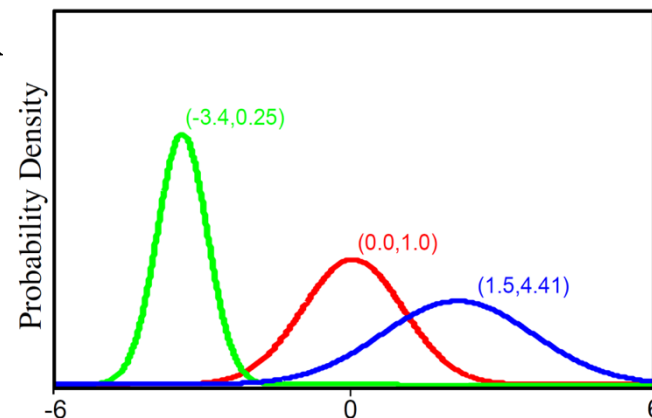


Image sources: "Pattern Recognition and Machine Learning", Christopher Bishop

"Computer Vision: Models, Learning, and Inference", Simon Prince

Multivariate Normal Distribution

- Multivariate normal distribution describes a **D -dimensional continuous variable** $\mathbf{X}$, i.e. $\mathbf{x} \in \mathbb{R}^D$.
- D -dimensional **mean** $\boldsymbol{\mu} \in \mathbb{R}^D$, and $D \times D$ symmetrical positive definite **covariance** matrix $\boldsymbol{\Sigma} \in \mathbb{R}_+^{D \times D}$.

$$p(\mathbf{X} = \mathbf{a} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{D/2} |\boldsymbol{\Sigma}|^{1/2}} \exp\{ -0.5(\mathbf{a} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{a} - \boldsymbol{\mu}) \}, \quad \mathbf{a} \in \mathbb{R}^D$$

Or

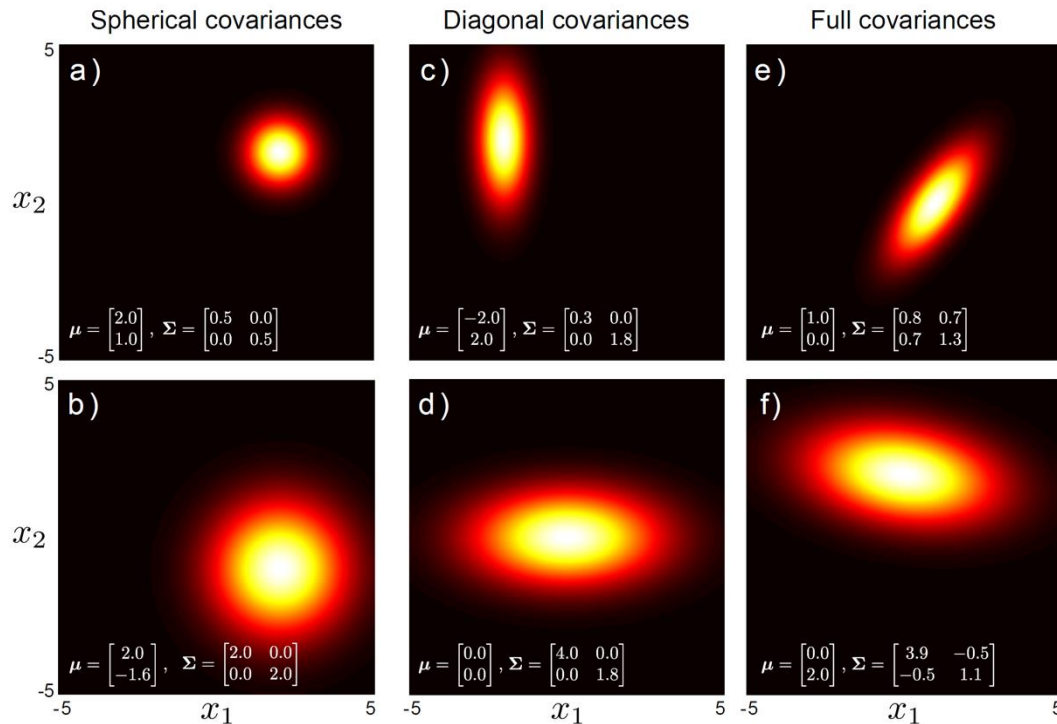
$$p(\mathbf{x}) = \frac{1}{(2\pi)^{D/2} |\boldsymbol{\Sigma}|^{1/2}} \exp\{ -0.5(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}) \}$$

$$p(\mathbf{x}) = \text{Norm}_x[\boldsymbol{\mu}, \boldsymbol{\Sigma}]$$

Types of Covariance

- Covariance matrix has three forms: **spherical**, **diagonal** and **full**.

$$\Sigma_{spher} = \begin{bmatrix} \sigma^2 & 0 \\ 0 & \sigma^2 \end{bmatrix} \quad \Sigma_{diag} = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix} \quad \Sigma_{full} = \begin{bmatrix} \sigma_{11}^2 & \sigma_{12}^2 \\ \sigma_{21}^2 & \sigma_{22}^2 \end{bmatrix}$$



Images Source: "Computer Vision: Models, Learning, and Inference", Simon Prince

Conjugate Distributions

- Conjugate distributions **model the parameters** of the probability distributions.
- **Product** of a probability distribution and its conjugate has the **same form** as the conjugate **times a constant**.
- Parameters of conjugate distributions are known as **hyperparameters** because they control the parameter distributions.

Distribution	Domain	Parameters modeled by
Bernoulli	$x \in \{0, 1\}$	beta
categorical	$x \in \{1, 2, \dots, K\}$	Dirichlet
univariate normal	$x \in \mathbb{R}$	normal inverse gamma
multivariate normal	$\mathbf{x} \in \mathbb{R}^k$	normal inverse Wishart

Importance of Conjugate Distributions

1. **Learning the parameters θ** of a probability distribution:

Recall the **Bayes' Rule**:

1. Choose prior that is conjugate to likelihood

$$p(\theta|x) = \frac{p(x|\theta)p(\theta)}{\int p(x|\theta)p(\theta)d\theta}$$

2. Implies that posterior must have **same form as** conjugate prior distribution, i.e. **closed-form**.

3. Posterior must be a distribution which implies that evidence **must equal constant κ** from conjugate relation.

Importance of Conjugate Distributions

2. Marginalizing over parameters:

- i. Chosen as **conjugate** to other term.

$$p(x^*|\mathbf{x}) = \int \underbrace{p(x^*|\theta)}_{\text{conjugate}} \underbrace{p(\theta|\mathbf{x})}_{\text{prior}} d\theta$$

- ii. Integral becomes easy --the product becomes a **constant times a distribution**.

Integral of constant times probability distribution
= constant times integral of probability distribution
= constant x 1 = constant

Importance of Conjugate Distributions

Proof:

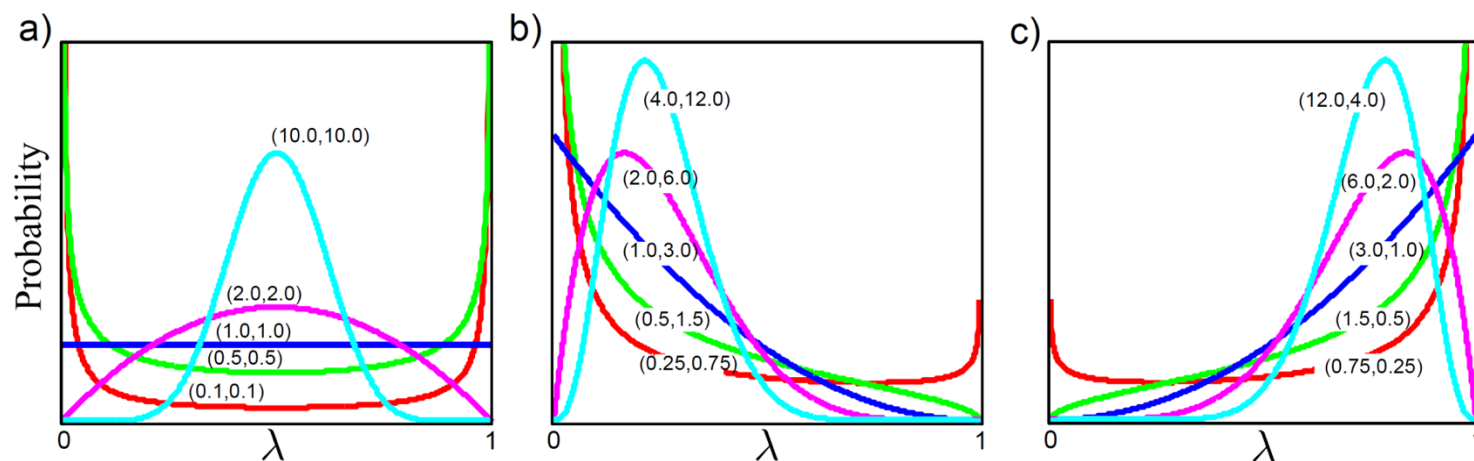
$$\begin{aligned} p(x^*|x) &= \frac{p(x^*, x)}{p(x)} && \text{(Conditional probability)} \\ &= \frac{\int p(x^*, x, \theta) d\theta}{p(x)} && \text{(Marginal probability)} \\ &= \frac{\int p(x^*, \theta|x) \cancel{p(x)} d\theta}{\cancel{p(x)}} && \text{(Conditional probability)} \\ &= \int p(x^*|x, \theta) p(\theta|x) d\theta && \text{(Conditional probability)} \\ &= \int p(x^*|\theta) p(\theta|x) d\theta && \text{(Conditional Independence)} \end{aligned}$$

Conjugate Distribution: Beta Distribution

- Conjugate distribution of **Bernoulli distribution**.
- Defined over parameter of the Bernoulli distribution $\lambda \in [0,1]$.

$$p(\lambda) = \frac{\Gamma[\alpha + \beta]}{\Gamma[\alpha]\Gamma[\beta]} \lambda^{\alpha-1}(1 - \lambda)^{\beta-1}$$

$$p(\lambda) = \text{Beta}_{\lambda}[\alpha, \beta]$$



Images Source: "Computer Vision: Models, Learning, and Inference", Simon Prince

Conjugate Distribution: Beta Distribution

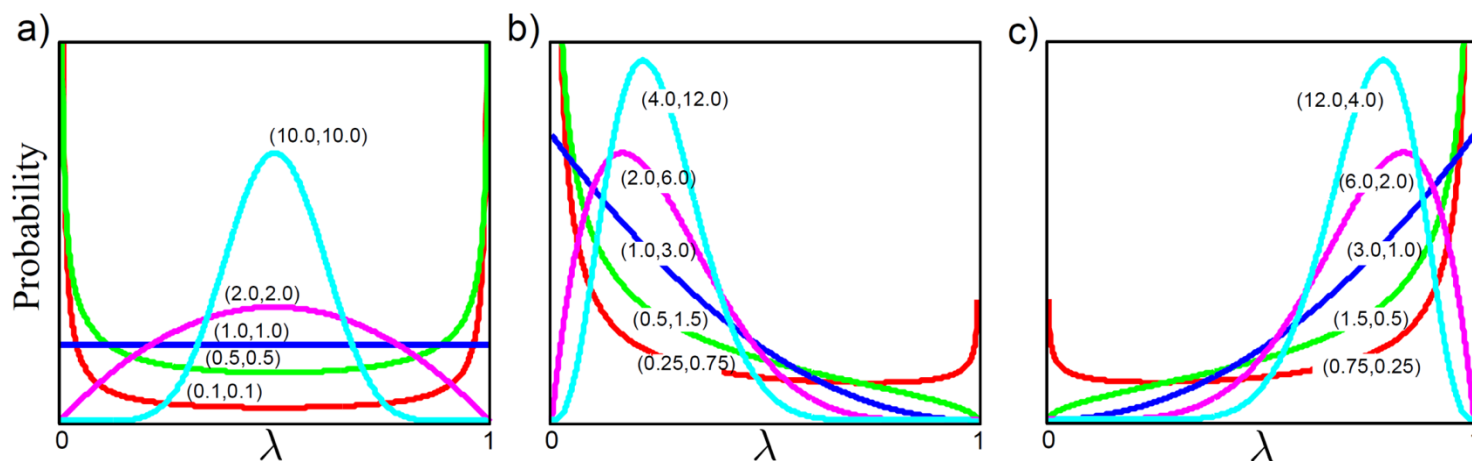
$$p(\lambda) = \frac{\Gamma[\alpha + \beta]}{\Gamma[\alpha]\Gamma[\beta]} \lambda^{\alpha-1} (1 - \lambda)^{\beta-1}$$
$$p(\lambda) = \text{Beta}_\lambda[\alpha, \beta]$$

Gamma Function:

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt, \quad z \in \mathbb{C}$$

$$\Gamma(n) = (n - 1)!, \quad n \in \mathbb{R}_{>0}$$

- **Two hyperparameters** $\alpha, \beta > 0$.



Images Source: "Computer Vision: Models, Learning, and Inference", Simon Prince

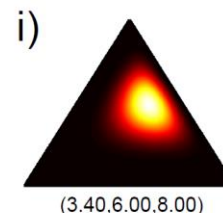
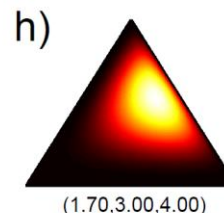
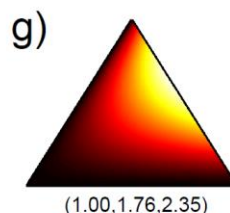
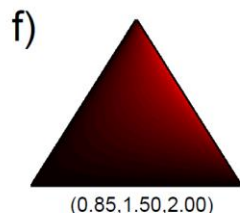
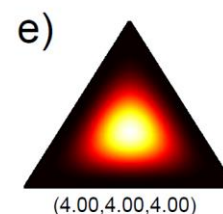
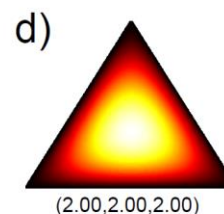
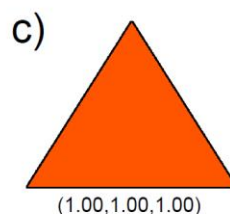
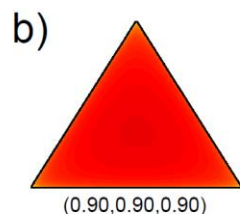
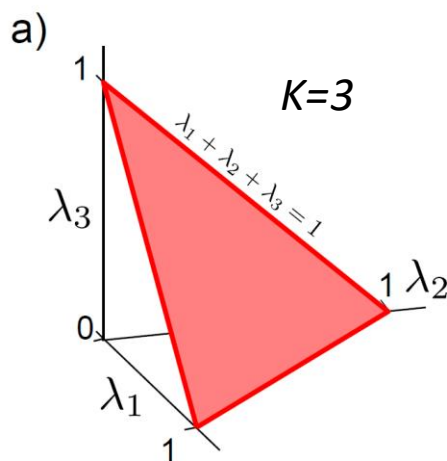
Conjugate Distribution: Dirichlet Distribution

- Conjugate distribution of **categorical distribution**.
- Defined over K parameters of Categorical distribution, $\lambda_k \in [0,1]$, where $\sum_k \lambda_k = 1$.

$$p(\lambda_1, \dots, \lambda_K) = \frac{\Gamma[\sum_{k=1}^K \alpha_k]}{\prod_{k=1}^K \Gamma[\alpha_k]} \prod_{k=1}^K \lambda_k^{\alpha_k - 1},$$
$$p(\lambda_1, \dots, \lambda_K) = \text{Dir}_{\lambda_1 \dots \lambda_K}[\alpha_1, \dots, \alpha_K]$$



Peter Gustav Lejeune Dirichlet
(1805-1859)

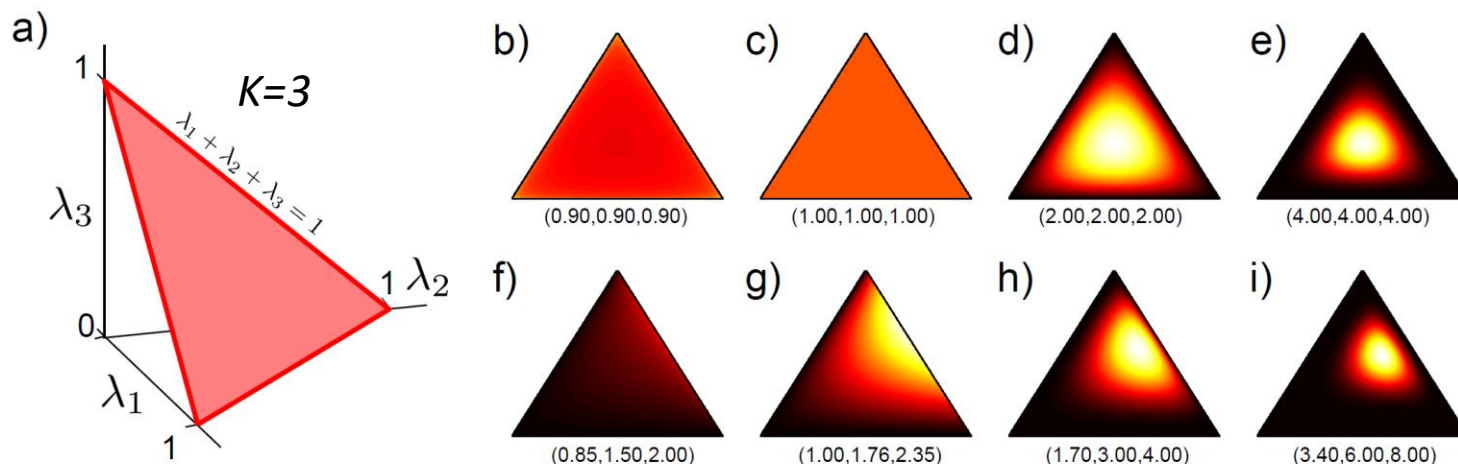


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Conjugate Distribution: Dirichlet Distribution

$$p(\lambda_1, \dots, \lambda_K) = \frac{\Gamma[\sum_{k=1}^K \alpha_k]}{\prod_{k=1}^K \Gamma[\alpha_k]} \prod_{k=1}^K \lambda_k^{\alpha_k - 1},$$
$$p(\lambda_1, \dots, \lambda_K) = \text{Dir}_{\lambda_1 \dots \lambda_K}[\alpha_1, \dots, \alpha_K]$$

- K hyperparameters $\alpha_k > 0$.



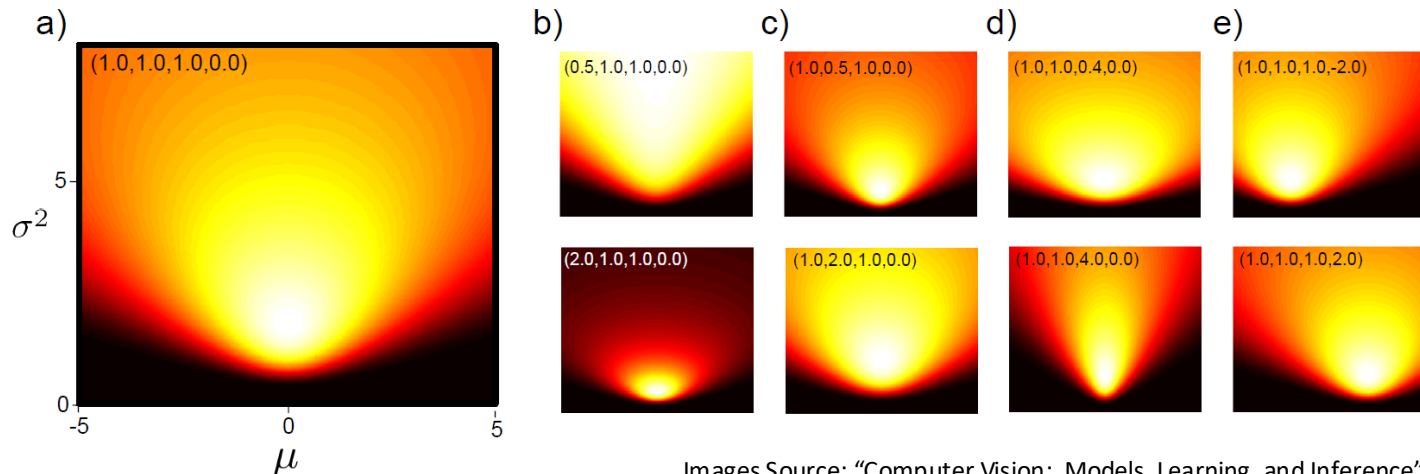
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Conjugate Distribution: Normal Inverse Gamma Distribution

- Conjugate distribution of **univariate normal distribution**.
- Defined on parameters $\mu, \sigma^2 > 0$ of univariate normal distribution.

$$p(\mu, \sigma^2) = \frac{\sqrt{\gamma}}{\sigma\sqrt{2\pi}} \frac{\beta^\alpha}{\Gamma[\alpha]} \left(\frac{1}{\sigma^2}\right)^{\alpha+1} \exp\left[-\frac{2\beta + \gamma(\delta - \mu)^2}{2\sigma^2}\right]$$

$$p(\mu, \sigma^2) = \text{NormInvGam}_{\mu, \sigma^2}[\alpha, \beta, \gamma, \delta]$$



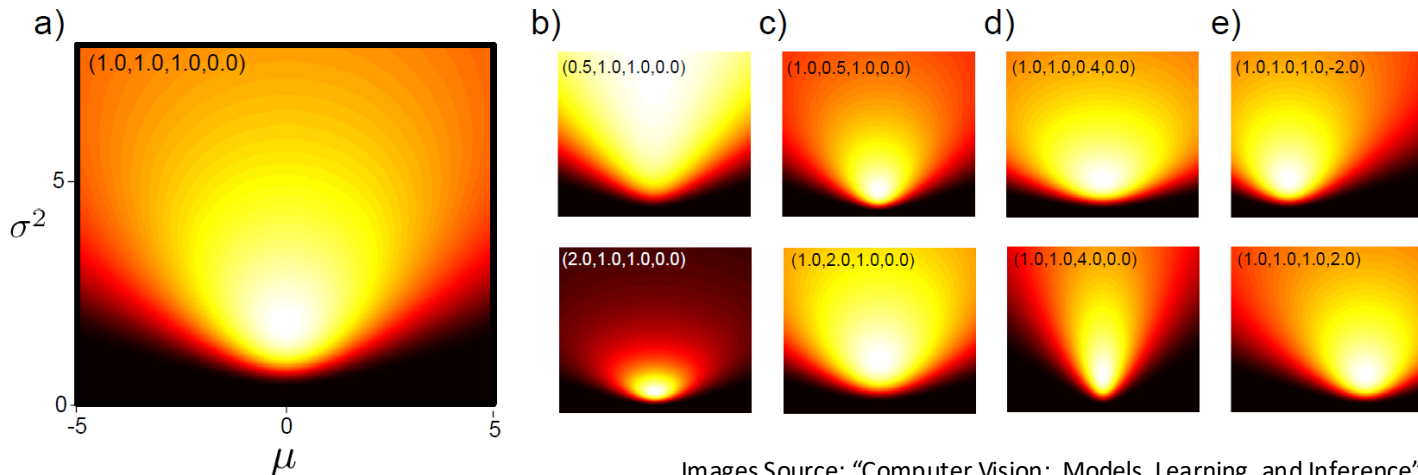
Images Source: "Computer Vision: Models, Learning, and Inference", Simon Prince

Conjugate Distribution: Normal Inverse Gamma Distribution

$$p(\mu, \sigma^2) = \frac{\sqrt{\gamma}}{\sigma\sqrt{2\pi}} \frac{\beta^\alpha}{\Gamma[\alpha]} \left(\frac{1}{\sigma^2}\right)^{\alpha+1} \exp\left[-\frac{2\beta + \gamma(\delta - \mu)^2}{2\sigma^2}\right]$$

$$p(\mu, \sigma^2) = \text{NormInvGam}_{\mu, \sigma^2}[\alpha, \beta, \gamma, \delta]$$

- **Four hyperparameters** $\alpha, \beta, \gamma > 0$ and $\delta \in \mathbb{R}$.



Images Source: "Computer Vision: Models, Learning, and Inference", Simon Prince

Conjugate Distribution: Normal Inverse Wishart



John Wishart
(1898-1956)

- Conjugate distribution of **multivariate normal distribution**.
- Defined on parameters $\boldsymbol{\mu}$, $\boldsymbol{\Sigma}$ of multivariate normal distribution.

$$p(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{\gamma^{D/2} |\boldsymbol{\Psi}|^{\alpha/2} \exp[-0.5 (\text{Tr} [\boldsymbol{\Psi} \boldsymbol{\Sigma}^{-1}] + \gamma (\boldsymbol{\mu} - \boldsymbol{\delta})^T \boldsymbol{\Sigma}^{-1} (\boldsymbol{\mu} - \boldsymbol{\delta}))]}{2^{\alpha D/2} (2\pi)^{D/2} |\boldsymbol{\Sigma}|^{(\alpha+D+2)/2} \Gamma_D[\alpha/2]}$$

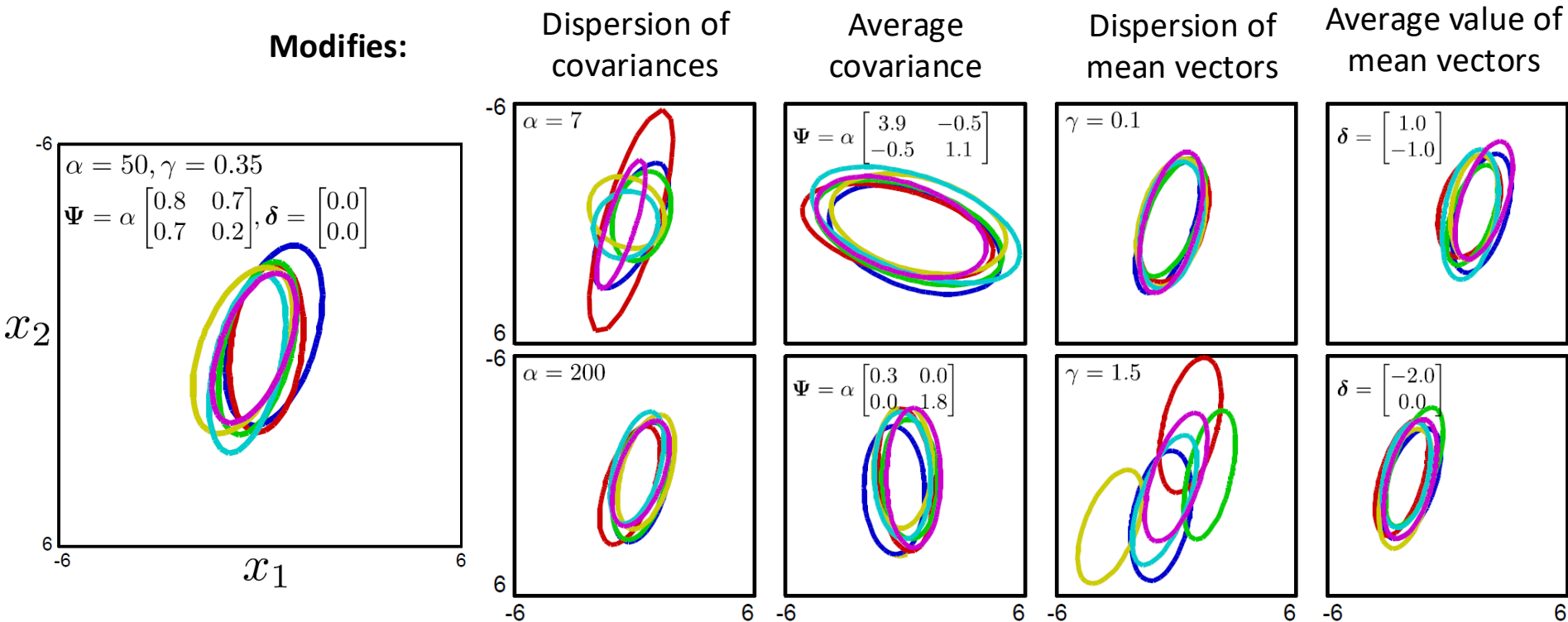
$$p(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \text{NorIWis}_{\boldsymbol{\mu}, \boldsymbol{\Sigma}}[\alpha, \boldsymbol{\Psi}, \gamma, \boldsymbol{\delta}]$$

- **Four hyperparameters**: a positive scalar α , a positive definite matrix $\boldsymbol{\Psi} \in \mathbb{R}_+^{D \times D}$, a positive scalar γ , and a vector $\boldsymbol{\delta} \in \mathbb{R}^D$.

Multivariate gamma function: $\Gamma_D[a] = \pi^{a(a-1)/4} \prod_{j=1}^a \Gamma[a + (1-j)/2]$

Conjugate Distribution: Normal Inverse Wishart

- Samples from Normal Inverse Wishart:



Images Source: "Computer Vision: Models, Learning, and Inference", Simon Prince

Example 1: Conjugate Distribution

Find: The **posterior distribution** of the parameter (μ, σ) from a univariate Gaussian distribution.

Solution: Assuming observations $x[i]$ are **independent and identically distributed** (i.i.d), we get:

$$p(\theta|x) = \frac{\prod_{i=1}^N p(x[i] | \theta)p(\theta)}{p(x)} = \frac{\prod_{i=1}^N p(x[i] | \theta)p(\theta)}{\int \prod_{i=1}^N p(x[i] | \theta)p(\theta) d\theta}$$

where:

$$\prod_{i=1}^N p(x[i] | \theta)p(\theta) = \prod_{i=1}^N \text{Norm}_{x[i]}[\mu, \sigma^2] \text{NormInvGam}_{\mu, \sigma^2}[\alpha, \beta, \gamma, \delta]$$

Example 1: Conjugate Distribution

We have

$$\prod_{i=1}^N p(x[i]|\theta)p(\theta) = \prod_{i=1}^N \text{Norm}_{x[i]}[\mu, \sigma^2] \text{NormInvGam}_{\mu, \sigma^2}[\alpha, \beta, \gamma, \delta]$$

Rearranging:

$$\prod_{i=1}^N p(x[i]|\theta)p(\theta) = \underbrace{\kappa[\alpha, \beta, \gamma, \delta, x]}_{\text{Constant}} \text{NormInvGam}_{\mu, \sigma^2}[\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}]$$

where

$$\tilde{\alpha} = \alpha + \frac{N}{2},$$

$$\tilde{\delta} = \frac{(\gamma\delta + \sum_i x[i])}{\gamma + N},$$

$$\tilde{\gamma} = \gamma + N,$$

$$\tilde{\beta} = \frac{\sum_i x[i]^2}{2} + \beta + \frac{\gamma\delta^2}{2} - \frac{(\gamma\delta + \sum_i x[i])^2}{2(\gamma + N)}.$$

Example 1: Conjugate Distribution

Putting into the Bayes' rule, we get:

$$p(\theta|x) = \frac{\prod_{i=1}^N p(x[i]|\theta)p(\theta)}{p(x)} = \frac{\prod_{i=1}^N p(x[i]|\theta)p(\theta)}{\int \prod_{i=1}^N p(x[i]|\theta)p(\theta) d\theta}$$

$$p(\theta|x) = \frac{\cancel{\kappa[\alpha, \beta, \gamma, \delta, x]} \text{NormInvGam}_{\mu, \sigma^2}[\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}]}{\cancel{\kappa[\alpha, \beta, \gamma, \delta, x]} \underbrace{\int \int \text{NormInvGam}_{\mu, \sigma^2}[\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}] d\mu d\sigma^2}_{=1}}$$

$$p(\theta|x) = \text{NormInvGam}_{\mu, \sigma^2}[\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}]$$

Same form as conjugate prior,
i.e., Normalized Inverse Gamma!

Example 2: Conjugate Distribution

Find: $p(x^*|\mathbf{x})$ for a univariate Gaussian.

Solution:

$$\begin{aligned} p(x^*|x) &= \int \int p(x^*|\mu, \sigma^2) p(\mu, \sigma^2|x) d\mu d\sigma^2 \\ &= \int \int \text{Norm}_{x^*}[\mu, \sigma^2] \text{NormInvGam}_{\mu, \sigma^2}[\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}] d\mu d\sigma^2 \\ &= \kappa[\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}, x^*] \underbrace{\int \int \text{NormInvGam}_{\mu, \sigma^2}[\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}] d\mu d\sigma^2}_{= 1} \\ &= \kappa[\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}, x^*] \end{aligned}$$

This is a constant (function of $\{x^*, x[1], \dots, x[N]\}$)!

Example 2: Conjugate Distribution

This is a constant (function of $\{x^*, x[1], \dots, x[N]\}$)!

$$p(x^*|x) = \kappa[\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}, x^*] = \frac{1}{\sqrt{2\pi}} \frac{\tilde{\beta}^{\tilde{\alpha}} \sqrt{\tilde{\gamma}}}{\check{\beta}^{\check{\alpha}} \sqrt{\check{\gamma}}} \frac{\Gamma[\check{\alpha}]}{\Gamma[\tilde{\alpha}]}$$

where

$$\begin{aligned}\check{\alpha} &= \tilde{\alpha} + 1/2, & \check{\gamma} &= \tilde{\gamma} + 1 \\ \check{\beta} &= \frac{x^{*2}}{2} + \tilde{\beta} + \frac{\tilde{\gamma}\tilde{\delta}^2}{2} - \frac{(\tilde{\gamma}\tilde{\delta} + x^*)^2}{2(\tilde{\gamma} + 1)}.\end{aligned}$$

Summary

You have learned how to:

1. Describe uncertain quantities with **random variables** and **joint probabilities**.
2. Explain the basic rules of probability – **sum**, **product**, **Bayes'**, **independence** and **expectation** rules.
3. Use the common probabilities distributions – **Bernoulli**, **categorical**, **univariate** and **multivariate normal** distributions.
4. Explain the use of **conjugate distributions**.